

REVIEW

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Emerging technologies for smart and sustainable precision agriculture

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Abstract

To maximize agricultural operations and ensure increased productivity, sustainability, and efficiency, precision agriculture makes use of innovative technologies. This review article examines the incorporation of artificial intelligence, machine learning, automation, cloud computing, and Internet of Things sensors as they relate to agrarian engineering. IoT sensors are essential to the collection of real-time data because they offer insightful information about crop status, weather patterns, and soil health. The efficient data processing, storage, and accessibility made possible by cloud computing make scalable and real-time analytics possible. While automation technologies, like robotics and autonomous systems, improve operational efficiency and save labor costs, AI and ML help with predictive analytics, illness diagnosis, and fertilization process optimization. This study showcases case studies that demonstrate how these technologies have been successfully implemented, emphasizing the observable advantages in terms of increased yield, reduced costs, and resource conservation. It also goes into the technological, financial, and privacy issues that come with combining various technologies, it provides information on existing constraints and possible workarounds. In addition, the assessment points out new directions for research, including the deployment of 5G, developments in edge computing, and the creation of increasingly complex AI algorithms.

Highlights

Integrates IoT sensors, cloud computing, AI, ML, and automation technologies in precision agriculture.

Reviews real-world implementations and case studies showcasing enhanced efficiency and sustainability.

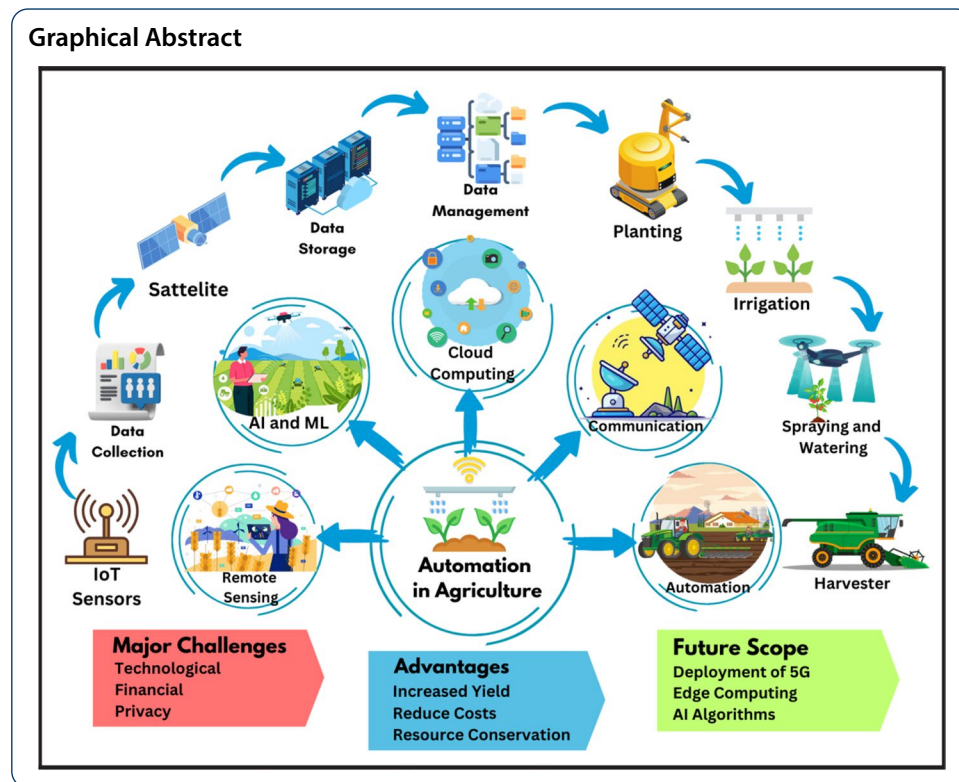
Discusses challenges such as data integration, scalability, and security.

Explores future directions, including emerging technologies and sustainability initiatives.

Provides comprehensive insights into the evolving landscape of precision agriculture.

Keywords Precision agriculture, IoT sensors, Cloud computing, Artificial intelligence, Automation in agriculture





1 Introduction

The application of innovative technologies to enhance and control agricultural productivity is known as precision agriculture, or PA. By precisely applying inputs like water, fertilizer, and pesticides on the basis of real-time data gathered from numerous sensors and other technology, this method more efficiently uses resources and increases agricultural yields [1]. PA seeks to improve farming businesses' profitability, sustainability, and production through the use of data-driven insights. It tackles several critical problems facing current agriculture, such as the need to feed a growing world population, and lessen the impact agriculture has on the environment [2] and increase farming's economic viability [3]. Through improved resource management and less waste, PA helps farmers make more informed decisions that eventually improve environmental sustainability and food security [4].

Over the past few decades, the idea of precision agriculture has undergone tremendous change. The invention of the global positioning system (GPS) in the 1980s, which made precise field mapping and positioning possible, is credited with giving rise to PA. In the early days of PA, yield mapping using GPS gave farmers insights into the geographical heterogeneity in their fields [5]. Developments in GIS drove the field of precision agriculture forward in the 1990s. GIS made it possible to integrate and analyze geographical data, which made field management more accurate. During this time, researchers also developed variable rate technology (VRT), which allowed for the application of inputs based on field variability at a specific site [6]. Significant technology developments in the 2000s increased PA's reach and potential. More regular and thorough information regarding crop health and field conditions became available with the widespread use of remote sensing technology and high-resolution satellite photography

[7, 8]. Simultaneously, the advancement of advanced modeling and data analytics technologies improved the capacity to understand and apply this data.

Precision agriculture has seen an evolution recently due to the convergence of artificial intelligence (AI), machine learning (ML), cloud computing, and the internet of things (IoT) [9]. IoT sensors make it possible to monitor several parameters in real-time, including temperature, nutrient levels, and soil moisture. AI and ML algorithms then evaluate this data to produce predictive analytics and actionable insights (Fig. 1). Large volumes of data can be processed, stored, and accessed more easily thanks to cloud computing, which helps farmers run their businesses more profitably [10]. All things considered, the development of precision agriculture is a reflection of both the ongoing progress in technology and the increasing awareness of the necessity of efficient and sustainable farming methods. The potential for PA to drastically alter farming methods and tackle global agricultural issues will only grow as these technologies advance.

The purpose of this paper is to provide a comprehensive analysis of the integration of automation, cloud computing, artificial intelligence, machine learning, and Internet of Things sensors with precision agriculture. This analysis includes an examination of contemporary technologies, their applications, and the benefits they offer to modern farming practices. The assessment will address the limitations and challenges of these technologies and offer recommendations for future research directions and potential solutions. This review's objectives are to define precision agriculture, look at its historical evolution, and evaluate IoT sensors in real-time data collection. It also examines how cloud computing can help with the maintenance of agricultural data, how AI and ML can be used to improve decision-making, and how automation technologies impact labor requirements and farming productivity. The study also discusses the challenges and limitations associated with the application of integrated technologies in precision agriculture, offers examples of effective implementations, and pinpoints areas in need of more investigation. The review aims to provide useful information about the transformative potential of contemporary technology in agriculture to guide academics, practitioners, and policymakers toward sustainable and productive farming practices.

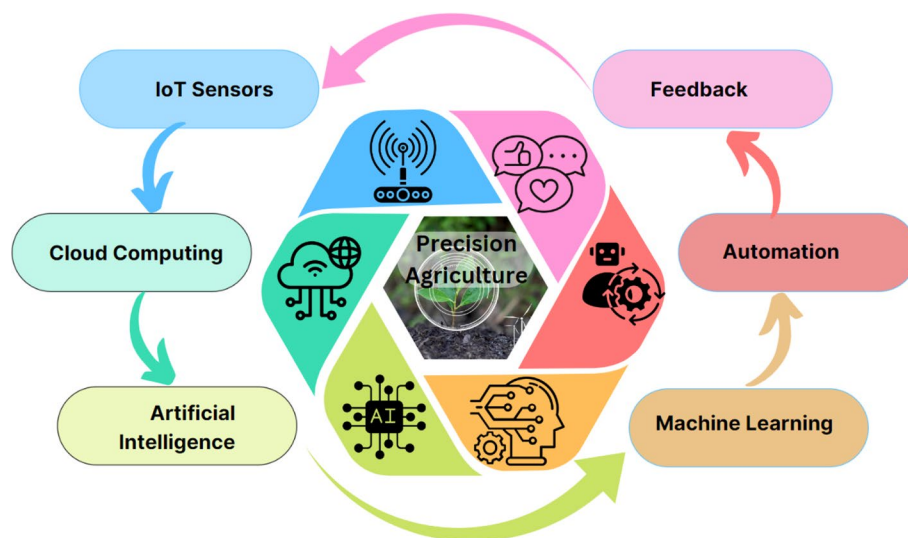


Fig. 1 Precision agriculture components

2 Review methodology

2.1 Review scope and literature identification

This review was carefully planned to look at how new digital technologies like the IoT, AI and ML, cloud computing, and automation are being used in agricultural engineering and precision farming. The primary objective was to evaluate how these technologies are applied, assess their individual and comparative effectiveness, and explore their combined potential to enhance sustainability, productivity, and real-time decision-making. The review included multiple technological domains, emphasizing the interconnectivity and collaborative application of these tools to deliver a comprehensive overview of the smart farming landscape.

To find relevant studies, a structured search was conducted across several scholarly databases, such as Scopus, IEEE Xplore, Web of Science, ScienceDirect, SpringerLink, and Google Scholar. It used a mix of targeted keywords like "Precision Agriculture," "IoT in Farming," "AI for Crop Monitoring," "Cloud Computing in Agriculture," "ML-based Irrigation," and "Autonomous Agricultural Vehicles" to do the literature search. Boolean operators were used to make the search results more specific (for example, "IoT AND Precision Farming" or "AI AND Pest Detection").

2.2 Screening, selection, and thematic organization

The database queries found an initial set of 432 articles. After getting rid of duplicate entries and checking the titles and abstracts for relevance, 151 articles were chosen for full-text review. The inclusion criteria emphasized peer-reviewed research articles, review papers, and pilot study reports that exhibited practical applications or suggested integrated frameworks for smart agriculture. Exclusion criteria removed sources that weren't peer-reviewed, articles that weren't in English, and articles that didn't go into enough technical or methodological detail. Of the 151 full-text articles examined, 112 studies were incorporated into the final synthesis due to their thematic depth and the quality of implementation data. Next, these studies were put into five main groups based on their themes: (1) sensor and IoT technologies for real-time monitoring, (2) AI and ML algorithms for prediction and control, (3) cloud and edge computing infrastructures for data handling, (4) automation and robotics for field operations, and (5) real-world case studies and comparisons. This thematic arrangement facilitated a systematic analysis of the distinct and collective functions of each technology, revealing performance trends and emphasizing interdependencies. The review aimed to achieve thorough coverage, but certain constraints were present. The selection process was limited to English-language publications, and certain industrial case studies, especially proprietary implementations, may not have been available for inclusion. This methodology provides a clear and repeatable way to bring together current research and uses of advanced technologies in agriculture.

2.3 Quantitative data handling and statistical considerations

This review does not include original experimentation but instead, it relies on statistical evaluation to compare findings from the existing studies. Quantitative metrics, including accuracy rates, percentage improvements, water savings, yield gains, and cost reductions, were extracted wherever available. To ensure consistency and comparability, standard metrics such as accuracy %, root mean square error (RMSE), R^2 values,

and percentage change were documented and analyzed across comparable applications, including AI-driven pest detection and IoT-based irrigation systems.

There was a lot of variability and uncertainty in the reported outcomes, especially when studies used different crops, environmental conditions, or sample sizes. To avoid over-generalization, the study presented the results as ranges or categorized them by region or technology type in these instances. While no original statistical tests (e.g., ANOVA, t-tests) were performed in this review, significance levels (e.g., $p < 0.05$) reported in the primary studies were noted and highlighted to underscore statistical robustness. Additionally, confidence intervals and standard deviations were recorded, where available, to reflect the degree of uncertainty and variation in the underlying data.

3 IoT Sensors in agriculture

3.1 Types of sensors

In precision agriculture, different kinds of IoT sensors are used to gather real-time data on different parts of the farming environment (Fig. 2). These sensors supply important information that helps make decisions based on data that can improve crop yields and profits [11].

Soil strength sensors: The soil strength sensor is a common type of IoT sensor used in farming. These sensors tell you a lot about the soil's structure and health by measuring its strength and compaction. Farmers can figure out the best times to plant, water, and harvest by looking at data from soil strength sensors. These sensors can also help farmers use more environmentally friendly methods by reducing soil erosion and nutrient runoff. [12].

Soil moisture sensors: The water content of the soil is measured using soil moisture sensors, which provide essential information for irrigation schedule optimization. By preventing both excessive and insufficient irrigation, these sensors help ensure that crops receive the proper quantity of water [13]. Time Domain Reflectometry (TDR) sensors, gypsum blocks, tensiometers, and capacitance sensors are a few examples of soil moisture sensor types. Depending on the type of soil and crop needs, each type has unique benefits and uses.

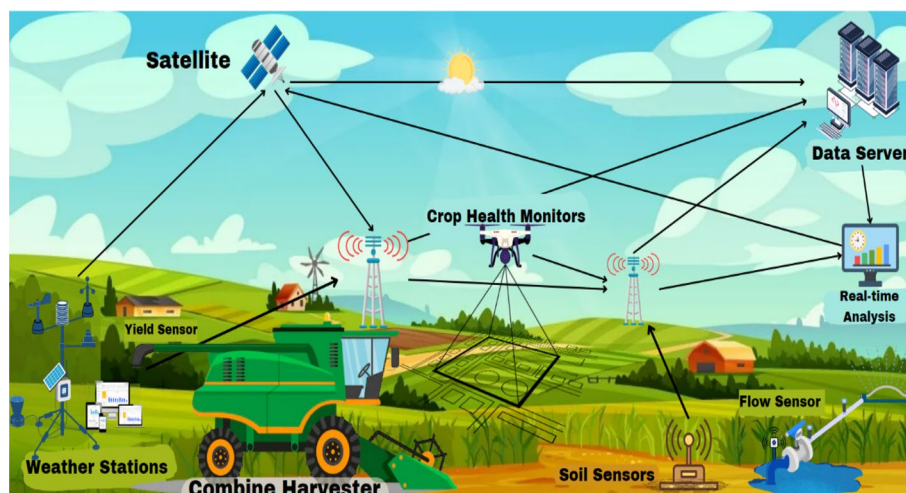


Fig. 2 Deployment of IoT sensors in a farm

Flow control and variable rate sensors: Variable rate sensors can increase irrigation efficiency during precision irrigation by adjusting the irrigation rate based on time and specific requirements. While spraying, the variable rate flow sensor, along with the pump and pressure sensors, operates on an IoT-based platform. This system effectively minimizes spray drift and chemical runoff, resulting in a significant reduction in chemical usage and improved environmental sustainability. These sensors are capable of efficiently controlling factors such as wind speed, direction, slope, and humidity, which directly impact the rate of flow and spray pressure [14].

Weather stations: IoT-enabled weather stations track a range of meteorological variables, including temperature, humidity, precipitation, wind speed, and sun radiation. Planning agricultural operations, controlling pest and disease risks, and forecasting weather patterns all depend on this data [15]. Several sensors are frequently combined into one modern weather station, giving thorough environmental data from a particular spot on the farm.

Crop health monitors: Crop health monitors analyze data such as leaf chlorophyll content, canopy temperature, and photosynthetic activity using optical sensors, such as multispectral and hyperspectral cameras. The sensors aid in the early diagnosis of illnesses, nutritional deficits, and insect infestations. Drones or UAVs (unmanned aerial vehicles) using these sensors are able to cover huge regions rapidly and provide accurate crop monitoring with high-resolution images and data.

Nutrient sensors: The nutrient sensors quantify the levels of essential nutrients such as potassium, phosphorus, and nitrogen in the soil. The sensors facilitate precise fertilizer delivery and provide information on soil fertility, thus promoting good nutrient management. Encouraging breakthroughs in precision agriculture technology are revealed by recent research into electrode materials for electrochemical monitoring of concentration in-situ soil fertilizer. Farmers can avoid over-fertilization, which may cause waste of resources and pollution of the environment, by using nutrient monitors. To provide crops with the nutrients they need at the right time, these sensors also help determine when to apply fertilizers.

All-terrain vehicle sensors: The rugged topography of hilly regions necessitates the use of all-terrain vehicles, particularly autonomous agricultural vehicles, which prove to be quite useful. Various slope sensors, GPS, and lidar are utilized to guarantee precise navigation and data acquisition for precision farming. These sensors assist farmers in maneuvering difficult terrain, guaranteeing that crops receive the essential nutrients and attention. Farmers can enhance efficiency and output in their fields by employing all-terrain vehicles fitted with sophisticated sensors. In general, these technical improvements are essential for encouraging sustainable agriculture practices and reducing environmental harm.

There are *gas sensors* that detect gases such as ammonia and carbon dioxide and are vital in explaining plant respiration and soil health. There are also *salinity sensors* that identify the levels of salt and *pH sensors* that monitor the acidity of the soil. Farmers can utilize the beneficial data provided by these sensors to make informed decisions about the health of their land and crops. Farmers can obtain a holistic insight into their agricultural activities and make appropriate changes to optimize vegetation growth and sustainability through the integration of different sensors. These advanced technologies make it possible for farmers to pursue more efficient and environmentally friendly

farming methods [16]. Table 1 summarizes different types of IoT sensors with specific uses.

3.2 Deployment strategies and network considerations

The implementation of IoT sensors in agricultural environments necessitates meticulous planning and evaluation of multiple aspects to guarantee dependable and effective data gathering and transfer.

Strategic placement of sensors: For the purpose of gathering representative data from various areas of the field, sensors must be positioned strategically. For example, to accommodate for variations in soil type and crop root zones, soil moisture sensors should be placed at different depths and locations [38]. It is imperative to install weather stations in open spaces that are unobstructed in order to guarantee precise measurement of meteorological parameters.

Wireless communication network: Typically, data from IoT sensors is wirelessly transmitted to a central system for analysis. Common wireless technologies for communication include cellular networks, Wi-Fi, Zigbee, and LoRaWAN. Power consumption, data rate, range, and environmental considerations are just a few examples of the factors that affect technology choice [39]. Because of its vast range and low power consumption, LoRaWAN is appropriate for big farms, whereas Wi-Fi would be chosen in smaller, more closely watched areas.

Power supply and energy management: It is imperative to guarantee a dependable source of electricity for sensors, particularly in isolated regions. IoT devices frequently use solar panels and battery-powered systems as sustainable energy sources [40]. Optimal data transmission protocols and energy-efficient sensors are necessary to prolong battery life and save on maintenance expenses.

Data integration and processing: To be processed and evaluated, the data gathered from several sensors needs to be connected to a single platform, which is frequently a cloud-based system. Standardization of data formats and interoperability across many sensor types are necessary for this [41]. For real-time applications, edge computing can also be used to process data locally, which lowers latency and bandwidth needs [42].

Scalability and network robustness: As farm activities grow, the sensor network should be scalable to allow for the addition of new sensors. To guarantee uninterrupted data transmission despite any interruptions, network resiliency is also crucial [43]. The sensor network can be made more reliable by including self-healing capabilities and redundant communication channels.

Farmers can effectively monitor and manage their agricultural operations with IoT sensors, improving productivity, sustainability, and efficiency by taking these deployment strategies and network considerations into account.

4 Cloud computing in agricultural data management

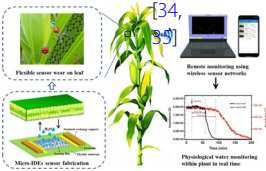
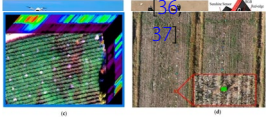
4.1 Data collection, storage, and processing

Since cloud computing offers reliable platforms for data processing, storage, and collection, it has completely changed the way data management is done in precision agriculture. Cloud-based systems can effectively manage the enormous volume of data produced by IoT sensors, drones, satellites, and other devices (Fig. 3).

Table 1 Types of IoT sensors used in agriculture

Sensor Type	Function	Applications	Setup/Schematic/Illustrative figures	References
Soil moisture sensor	Measures soil moisture levels	Irrigation management, drought monitoring		[17]
Weather station	Monitors weather conditions	Climate monitoring, weather forecasting		[18, 19]
Crop health monitor	Detects crop health and stress	Disease detection, nutrient deficiency analysis		[20, 21]
pH sensor	Measures soil acidity/alkalinity	Soil management, fertilizer application		[22, 23]
Light sensor	Measures light intensity	Crop growth monitoring, greenhouse management		[24, 25]
Temperature sensor	Measures ambient temperature	Climate control, frost protection		[26, 27]
Humidity sensor	Measures air moisture levels	Greenhouse management, pest control		[28, 29]
CO ₂ sensor	Measures carbon dioxide levels	Photosynthesis optimization, greenhouse monitoring		[30, 31]
Nutrient sensor	Measures nutrient levels in soil	Fertilizer application, soil health assessment		[32, 33]

Table 1 (continued)

Sensor Type	Function	Applications	Setup/Schematic/Illustrative figures	References
Leaf wetness sensor	Detects moisture on leaf surface	Disease prediction, irrigation scheduling		[34, 35]
Multi-spectral sensor	Captures various light spectra	Crop health monitoring, vegetation analysis, crop yield monitoring		[36, 37]

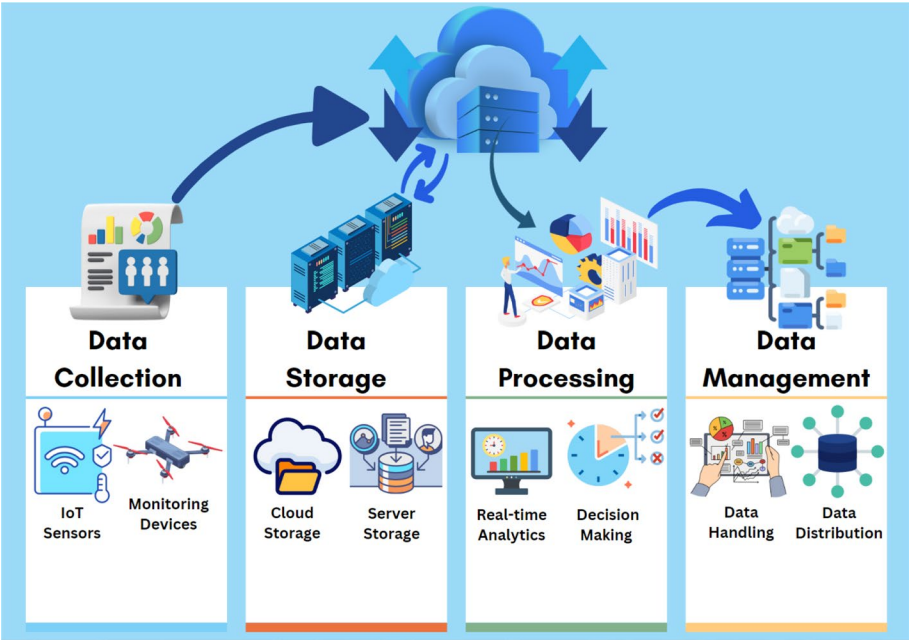


Fig. 3 Cloud-based agricultural data management system

Data collection: Field-installed IoT sensors gather data continually on a range of features, including temperature, humidity, crop health, and weather. The real-time wireless transmission of this data to cloud servers is accomplished [44]. Drones and satellite photography offer high-resolution spatial data that may be linked to the cloud to enable an in-depth understanding of crop conditions and variability in the field.

Data storage: Scalable storage solutions provided by cloud systems can handle large data volumes. For agricultural operations that eventually produce terabytes of data, this adaptability is crucial [45]. Data repositories that are safe, redundant, and easily accessible can be found with storage systems like Microsoft Azure, Amazon S3, and Google Cloud Storage. These services guarantee data availability and integrity even in the case of hardware malfunctions.

Data processing: Cloud computing makes it possible to transform unprocessed data into insightful knowledge. The gathered data is analyzed using big data tools, ML

algorithms, and advanced analytics [46]. Real-time data processing capabilities allow for the immediate interpretation of sensor data, which enables timely decision-making. Predictive models, for example, can forecast insect outbreaks and assess weather data, allowing for proactive actions [47].

4.2 Integrated system architecture for smart farming

Digital technology system architecture begins with IoT sensors measuring environmental factors, including soil water, temperature, humidity, and the presence of pests. Data is sent to a cloud server or edge computing nodes based on the latency sensitivity of the application. Cloud computing facilitates the aggregation of big data and long-term training of models with AI and ML algorithms.

AI and ML models analyze collected data to generate insights, like crop diseases, yield forecasting, and irrigation routines. AI/ML model output initiates decisions for intelligent agricultural equipment like tractors and sprayers. The entire system is remotely monitored, allowing farmers to oversee operations and adjust parameters in real-time. This design moves away from manual-based processes towards a data-driven, autonomous, and scalable environment, boosting productivity, resource utilization, and long-term viability. Figure 4 presents an integrated architecture model where IoT sensors collect real-time field data, which is transmitted to the cloud and edge nodes for processing. AI and ML algorithms interpret the data to produce actionable recommendations, which then guide automated responses such as variable-rate spraying or autonomous machinery navigation.

4.3 Benefits of cloud-based solutions for scalability and real-time analytics

There are several advantages to using cloud-based solutions in agriculture, especially when it comes to scalability and real-time analytics.

Scalability: Storage and processing power are essentially limitless with cloud computing. For agricultural operations of all sizes, from tiny farms to massive agribusinesses, this scalability is essential [48]. When their need for data increases, farmers can scale up their cloud service investments from a small initial outlay. This idea also reduces upfront costs by making pay-as-you-go resource management possible.

Real-time analytics: Performing analytics in real-time is one of cloud computing's major benefits. Farmers can receive up-to-date information on crop status and environmental factors by using real-time processing and analysis of data from IoT sensors [49]. Real-time analytics make it possible for precision agriculture practices like variable rate

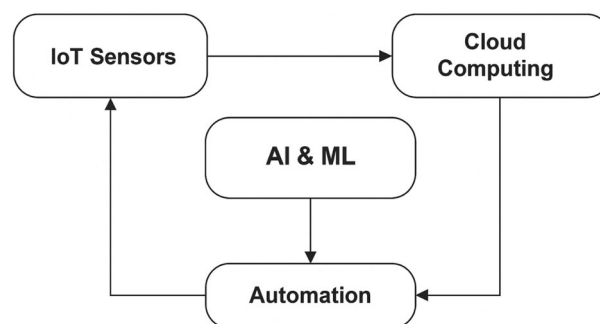


Fig. 4 Architecture of technology integration

input application (for example, water, fertilizer, and pesticides). Real-time data on soil moisture, for instance, can be used to program irrigation systems to supply water exactly where and when it is needed, maximizing water use and enhancing crop health [50].

Data integration and interoperability: Cloud platforms make it simpler to integrate data from various sources, such as IoT devices, drones, satellite imaging, and manual inputs. This integration provides a thorough understanding of the agricultural environment, which improves decision-making processes [51]. Standardized data formats and APIs enable easy data interchange and collaboration across diverse stakeholders in the agricultural value chain by promoting interoperability between various devices and systems.

Cost efficiency and accessibility: Cloud-based solutions reduce the need for on-premises hardware and maintenance, lowering operational costs. Additionally, cloud services often include robust security measures, protecting data from cyber threats and physical damage [52]. Data stored in the cloud can be accessed from anywhere with an internet connection, providing farmers with the flexibility to monitor and manage their operations remotely. This accessibility is particularly beneficial for large farms and operations spread across multiple locations.

Enhanced collaboration and knowledge sharing: Because cloud platforms enable several people to access and operate on the same data sets, they facilitate collaborative operations. Farmers, agronomists, and researchers can collaborate to solve agricultural problems by exchanging knowledge and best practices [52]. Cloud-hosted knowledge-sharing platforms facilitate the rapid adoption of innovative agricultural methods by disseminating research findings and technical improvements.

Because it provides scalable, effective, and real-time solutions that improve precision agriculture techniques, cloud computing is essential to the administration of agricultural data. Farmers may enhance resource management, boost production, and guarantee sustainability in their operations by utilizing cloud-based solutions.

5 Artificial intelligence and machine learning applications

5.1 Predictive analytics for crop yield estimation

In order to precisely predict crop yields, predictive analytics in agriculture makes use of ML and AI. In order to forecast crop performance in the future, these technologies examine historical data, weather patterns, soil characteristics, and other environmental parameters (Fig. 5).

Data collection and model training: Numerous sources, such as satellite imaging, IoT sensors, weather stations, and historical crop yield records, are used to gather large datasets. These datasets provide the foundation for training ML models [53]. Using this data, ML algorithms such as neural networks, decision trees, and regression models are trained to find patterns and correlations between crop yields and input variables [54].

Application and benefits: The use of ML and AI vision technologies in all-terrain vehicles for precision agriculture demonstrates how important they are to the automation of farm activities. Once trained, these models can forecast crop yields in various circumstances, assisting farmers in making well-informed choices regarding irrigation, fertilization, and planting. For instance, a model can forecast lower yields in particular field locations, resulting in focused interventions to enhance results [55]. Predictive analytics lowers waste, increases total agricultural productivity, and facilitates more effective

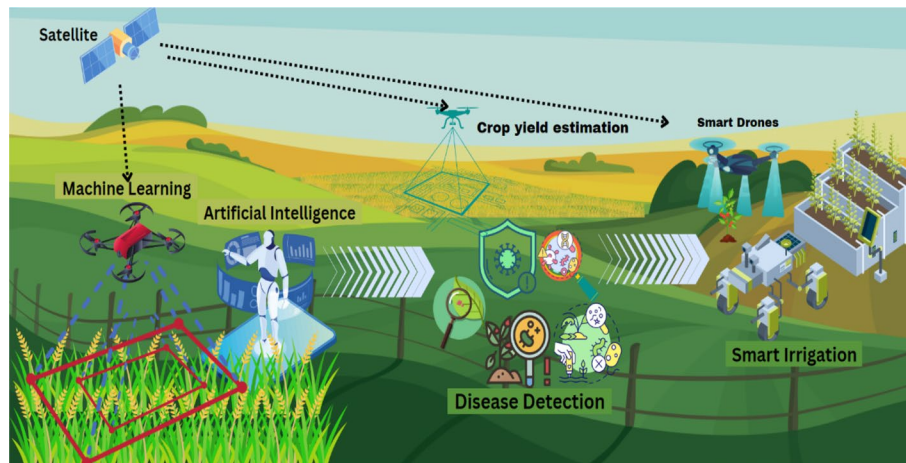


Fig. 5 AI and machine learning in precision agriculture

resource allocation. Precise forecasts of yield additionally aid in the organization of market tactics and logistics, guaranteeing increased profits for agricultural producers [56].

5.2 Disease and pest detection through image processing

Preventing major crop losses requires early detection of pests and plant diseases, which is why AI and ML are being utilized more and more in this regard. AI and ML algorithms can rapidly detect symptoms of illness or insect infestation by examining photos of crops, allowing farmers to respond swiftly to minimize harm (Fig. 5). By preventing crop devastation, this proactive strategy also lessens the need for excessive pesticide use, encouraging more environmentally friendly farming methods. Farmers can more successfully protect their crops and eventually increase their total output and profitability with the use of AI and ML technologies.

Image acquisition and preprocessing: Drones, satellites, or mobile equipment are used to take high-resolution photos of crops. After that, these photos undergo preprocessing to improve their characteristics and get rid of noise [57]. To make sure the photos are appropriate for ML algorithms to analyze, methods like image segmentation and normalization are used.

Machine learning algorithms for detection: For image-based disease and pest detection, convolutional neural networks (CNNs) and other deep learning models are frequently utilized [58]. The labeled datasets with pictures of both healthy and sick plants are used to train these algorithms [59]. By examining the patterns and characteristics in the photos, the trained models can recognize particular illness symptoms and insect infestations. CNNs can identify early indicators of rust or leaf blight in crops, enabling prompt intervention [60].

Application and benefits: Farmers can receive real-time notifications and continuous crop monitoring thanks to automated disease and pest detection systems. The early warning system aids in the application of targeted therapies, the prevention of disease transmission, and the reduction of pesticide use. These methods lower expenses related to broad-spectrum pesticide treatments and manual scouting while also improving crop health management and yields.

5.3 Smart irrigation systems using ML algorithms

Smart irrigation systems use AI and ML to optimize water usage, ensuring that crops receive the right amount of water at the right time.

Data integration and model development: For training ML models, data from soil moisture sensors, weather forecasts, evapotranspiration rates, and crop growth phases are combined. According to Togneri et al. (2022), these models forecast the best irrigation schedules depending on the existing and anticipated conditions. To evaluate this data and produce accurate irrigation suggestions, algorithms like neural networks, support vector machines (SVM), and random forests are used [61].

Real-time monitoring and control: Irrigation systems with intelligence constantly track soil moisture content and meteorological variables. Using this real-time data, the ML models dynamically modify irrigation plans to guarantee optimal water consumption [62]. By integrating these technologies with automated irrigation infrastructure, water application across various field zones may be precisely controlled [63].

Application and benefits: Smart systems optimize irrigation schedules to minimize water waste and enhance crop health. They make sure that crops receive the proper amount of water, which can greatly improve both productivity and quality [64]. Since these systems encourage sustainable water management techniques, they are especially helpful in areas where water is scarce. They also lower labor expenses related to managing irrigation manually [65].

Applications of AI and ML in agriculture provide substantial advantages, ranging from disease detection and smart irrigation systems to predictive analytics for crop production estimation, as well as in healthcare and maintaining smart cities [66]. They support sustainable farming methods, increase resource efficiency, and improve decision-making.

6 Automation technologies

6.1 Robotics and autonomous vehicles in farming

By automating tedious jobs and enhancing accuracy and efficiency across a range of agricultural activities, robotics and autonomous vehicles are transforming farming operations. A thorough analysis of the use of autonomous all-terrain vehicles in precision farming has shown how their increased automation and efficiency could help guide agriculture into the future [67]. The use of 3D printing in the design of all-terrain vehicles has accelerated advances in precision agriculture and opened up new opportunities for farming applications [68].

6.1.1 Types of robotics and autonomous vehicles

Drones and unmanned aerial vehicles (UAVs): UAVs with cameras and sensors are used for crop monitoring, aerial photography, and pesticide or fertilizer application. Drones enable farmers to make focused interventions by giving them access to real-time data on pest infestations, soil moisture levels, and crop health [69].

Autonomous tractors and machinery: Equipped with GPS, sensors, and computer vision systems, these vehicles can traverse fields, carry out precise tasks like planting, spraying, and harvesting, and plan routes to reduce fuel consumption and soil compaction [70].

Robotic harvesters and weeder: Specialized robots with robotic arms and cameras harvest fruits, vegetables, and other crops. Through image processing techniques, these robots are able to recognize ripe fruit, pick it gently without harming the crops, and get rid of weeds without using herbicides.

6.1.2 Advantages of robotics and autonomous vehicles

Enhanced productivity and efficiency: Farmers may concentrate on more strategic endeavors thanks to automation, which lowers labor expenses and the amount of time spent on manual labor. Autonomous vehicles can operate continuously, hence optimizing productivity throughout crucial periods like planting and harvesting seasons. An extensive analysis of the uses and advantages of autonomous agricultural vehicles reveals that they may significantly boost the productivity of hill farming [71].

Precision and accuracy: Robotics and autonomous vehicles are capable of carrying out tasks with a high degree of precision and accuracy, which results in consistent crop coverage and effective resource utilization. This accuracy reduces waste and raises crop quality in general [72].

Safety and sustainability: Automation helps to improve occupational safety by lowering the amount of chemicals and hazardous circumstances that farmworkers are exposed to. To increase accuracy in the spraying of agricultural treatments, an IoT-based semi-autonomous vehicle sprayer has been developed, demonstrating the convergence of IoT and automation in farming [73]. Additionally, automation reduces chemical runoff and soil compaction by permitting targeted applications of inputs like pesticides and fertilizers, which supports environmental sustainability [74].

6.2 Automated harvesting systems

Automated harvesting systems use AI, robotics, and computer vision to increase crop yields and speed up the harvesting process (Fig. 6).

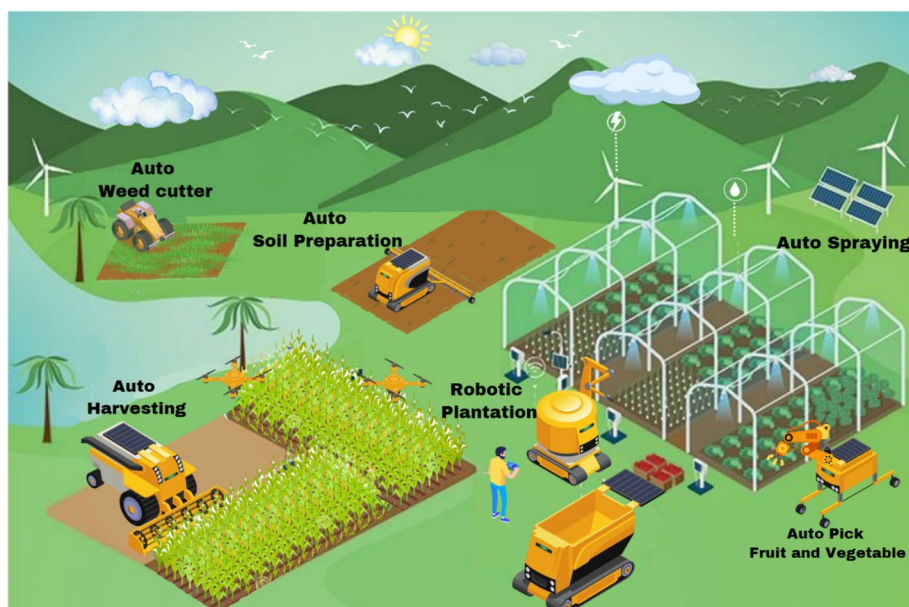


Fig. 6 Automated farming technologies

6.2.1 Technological components

Computer vision and machine learning: Computer Vision and ML facilitate the ability of machines to recognize mature fruits or vegetables, evaluate their superiority, and determine the most effective method for harvesting. AI algorithms utilize image analysis from camera captures to make immediate determinations regarding the selection and handling of harvested products [53].

Robotic arms and grippers: Automated harvesting devices are fitted with robotic prosthetics and specialized clamps designed to delicately collect fruits or vegetables without inflicting harm. These robotic devices imitate human dexterity and adjust to differences in crop size, shape, and ripeness [75].

6.2.2 Applications and benefits

Efficiency and cost savings: Automated harvesting systems reduce the need for manual labor, which is sometimes expensive and susceptible to labor scarcity. Through the automation of repetitive processes, these technologies enhance the efficiency of harvesting and decrease operational expenses [76].

Enhanced quality and yield: Automated harvesting machines can operate continuously without experiencing exhaustion, guaranteeing the timely collection of crops when they are at their optimal level of maturity. As a consequence, this leads to the production of superior-quality crops that have an extended period of freshness and enhanced appeal to potential buyers. In addition, automation reduces crop damage during harvesting, hence improving overall production [77].

Scalability and adaptability are key features of automated harvesting systems, as they can be easily adjusted to accommodate various crop types and sizes. This includes row crops such as lettuce and strawberries, as well as tree fruits like apples and oranges. These systems can adjust to different agricultural conditions and can be tailored to fulfill specific harvesting needs [78].

6.3 Integration of AI-driven decision-making tools

The integration of AI-driven decision-making tools, which optimize farming practices and improve agricultural outcomes, makes automation technologies possible [79].

Data-driven insights: AI systems process extensive data obtained from sensors, drones, and several other sources to offer practical and useful insights for farmers. The findings comprise suggestions for optimal planting schedules, effective irrigation management, strategic pest control methods, and well-structured crop rotation programs [80]. AI-driven decision-making technologies utilize historical data and real-time observations to forecast future trends, foresee hazards, and optimize resource allocation. This leads to more knowledgeable and strategic decision-making.

Integration with automation technologies: AI decision-making tools can be combined with robotics, autonomous vehicles, and automated systems to improve their functionality and efficiency. AI algorithms can enhance the efficiency of field operations by optimizing the routes of autonomous tractors using data on soil moisture levels and crop health [81]. These solutions facilitate the ability of automated technology to adjust to fluctuating climatic circumstances and dynamic farming situations in real-time, thereby optimizing productivity and reducing waste [82]. By utilizing these technologies, farmers may maximize resource utilization, minimize labor expenses, and enhance crop quality

and yields. Table 2 presents a comparison of automation technologies in agriculture, such as robotics for planting and harvesting, autonomous vehicles for soil preparation, AI-driven decision aids, drones, and smart greenhouses. It focuses on their applications, advantages, and limits in modern farming operations.

7 Real-world implementations

7.1 Application of integrated systems

To validate the effectiveness of integrated digital technologies in precision agriculture, numerous real-world deployments and pilot studies have been conducted across diverse geographical regions and operational contexts. These studies highlight both the tangible benefits and the practical challenges associated with field-level implementation.

A study in Quang Tho, Vietnam, utilized an IoT-based system integrated with LoRaWAN, ML and Augmented Reality (AR) to monitor environmental parameters such as temperature, soil moisture, and air humidity. This system not only improved farm management efficiency but also reduced the risk of crop failure [94]. Similarly, another study demonstrated the effectiveness of IoT-based smart agricultural monitoring systems using Wireless Sensor Networks (WSNs) and predictive analytics. This system achieved a 40% improvement in resource usage and a 30% increase in crop yield compared to traditional methods [95].

A study in India highlighted the use of AI models such as Decision Trees, SVM, and Neural Networks, combined with IoT sensors, to provide real-time insights into soil health, weather conditions, and crop growth. These technologies led to increased productivity and reduced costs for farmers [96]. Another study emphasized the role of AI in optimizing irrigation, fertilization, and pest control, with case studies demonstrating significant yield improvements and cost reductions [97].

Drones equipped with multispectral and hyperspectral sensors have become essential tools for crop monitoring and management. A comprehensive review highlighted the

Table 2 Comparison of automation technologies in agriculture

Technology	Applications	Benefits	Limitations	References
Robotics	Planting, weeding, harvesting	Increases efficiency, reduces labor	High initial cost, technical complexity	[83]
Autonomous vehicles	Soil preparation, seeding, spraying	Precision operations reduce errors	Requires advanced infrastructure	[84]
Automated harvesting systems	Fruit picking, vegetable harvesting	Increases speed, reduces waste	Limited adaptability to crop variations	[75]
AI-driven decision tools	Farm management, crop monitoring	Data-driven insights, improved decisions	Data privacy concerns, reliance on data quality	[85, 86]
Drones	Crop monitoring, spraying, mapping	High precision, broad coverage	Limited flight time, regulatory issues	[87]
Automated irrigation systems	Precision irrigation, water management	Water savings, increased yields	Initial setup cost, maintenance	[88]
Autonomous soil sampling	Soil analysis, nutrient mapping	Detailed soil data, time savings	Equipment cost, operational complexity	[89]
GPS-guided tractors	Plowing, planting, spraying	Reduced overlap, increased efficiency	High cost, dependency on GPS accuracy	[90]
Smart greenhouses	Climate control, resource management	Optimal growing conditions, efficiency	High setup and operational costs	[91]
Automated live-stock feeding	Timely and precise feeding	Improved animal health, reduced waste	Equipment cost, system reliability	[92]
Autonomous pest control	Targeted pest management	Reduced chemical use, environmental benefits	High initial investment, technical know-how	[93]

effectiveness of drones in various applications, including crop monitoring, multispectral mapping, precision spraying, and drone-assisted seeding. These technologies have demonstrated significant improvements in early disease detection, yield estimation, and irrigation management [98]. Drones have been successfully used in viticulture and cereal cultivation, showcasing their versatility and effectiveness in different agricultural contexts.

Digital Twin (DT) technology has emerged as a transformative solution for precision agriculture. By creating virtual replicas of farms, DTs enable data-driven decision-making, optimize crop management, and enhance soil health. A study highlighted the successful deployment of DTs in various regions, demonstrating their potential to improve agricultural efficiency, sustainability, and resilience [99]. Another study emphasized the role of DTs in replicating crop functionalities, supporting decision-making, and increasing water, fertilizer, and pesticide savings [100].

In Greece a study examined the integration of technology in agricultural practices towards agricultural sustainability. The results showed a significant positive effect of technology integration on agricultural sustainability, with cutting-edge technologies like AI, IoT, and precision agriculture improving resource efficiency, lowering environmental effects, and raising agricultural yields. The study emphasized the significance of infrastructure investment, supporting legislation, and farmer education in facilitating the adoption of agricultural technology [101].

In Italy a study evaluated the impacts of precision agriculture technologies on farm economic, agronomic, and environmental management. The results suggested that the implementation of precision technologies can guarantee economic and agri-environmental efficiency. The study focused on two farms cultivating durum wheat in central Italy over the period 2014–2022, demonstrating the potential of precision agriculture to contribute to sustainable development [102].

A case study in Dasheng Town, Yubei District, Chongqing, explored the enabling role of digital technology in the high-quality development of agriculture. The study summarized the positive impacts of digital technology on improving agricultural production efficiency, enhancing the quality of agricultural products, and promoting agricultural sustainability. The successful experience of the Dasheng Town smart agriculture project provided valuable insights for other regions to use digital technology to promote agricultural modernization [103].

In a study conducted in the Ubahu Community, Nigeria, the potential of precision agriculture technologies to transform agriculture is uncovered [104]. These technologies, such as biotechnology, nanotechnology, bioinformatics, blockchain, 3D printing, drones, robotics, sensors, IoT, and mobile technology, have the potential to decrease labor intensity, enhance agricultural input efficiency, control weeds, pests, and diseases, maximize transportation and storage systems, and improve farm record keeping and security arrangements. Impediments such as low incomes, poor infrastructure, poor awareness, poor government support, and fear of failure prevent their uptake. The study recommended measures such as increasing awareness, offering financial incentives, improving rural infrastructure and security, and organizing training programs to overcome these barriers.

John Deere, a prominent producer of farming equipment, provides precision agriculture solutions that incorporate IoT sensors, GPS technology, and cloud-based analytics.

Their technologies help farmers monitor field conditions, remotely manage equipment, and optimize planting and harvesting processes [105]. Farmers have experienced substantial enhancements in crop output, decreased expenses for inputs, and improved utilization of resources as a result of utilizing John Deere's integrated systems. Real-time data insights facilitate timely decision-making, which improves crop management and increases farm profitability [106].

Blue River Technology, a division of John Deere, developed the See & Spray precision sprayer. This advanced device utilizes computer vision and ML algorithms to instantly detect weeds and accurately administer herbicides only to the affected areas [107]. This focused strategy decreases pesticide consumption by up to 90% in comparison to conventional indiscriminate spraying techniques [108]. Farmers who have implemented Blue River's See & Spray technology have experienced notable reductions in herbicide expenses, minimized environmental harm, and enhanced effectiveness in weed management. The method enhances efficiency by obviating the necessity for human weed scouting and mitigating the proliferation of herbicide-resistant weeds [109].

FarmBot is an open-source, automated farming system that integrates robots, the IoT, and AI to provide small-scale, exceptionally efficient agriculture. The system comprises a robotic gantry that is outfitted with planting, watering, and weeding equipment. It may be operated remotely through a web-based interface [110]. FarmBot users have experienced enhanced productivity in domestic gardens and small-scale farms as a result of accurate planting and watering techniques, resulting in higher crop yields and less water usage. The system's ability to scale and its cost-effectiveness make it easily accessible to people and communities interested in sustainable agriculture [111].

Figure 7 illustrates the implementation of precision agriculture technologies in agricultural practices. Figure 7a shows the positive impact of GPS-guided tractors and automated tillage equipment on soil preparation, compaction reduction, and crop yield improvement. IoT sensors and automated sprayers minimize waste and environmental harm (Fig. 7b), while precision planting maximizes seed germination and crop growth (Fig. 7c). Automated harvesters and yield monitoring systems guarantee low crop loss and superior quality output, resulting in reduced labor costs and enhanced harvest efficiency (Fig. 7d). These technologies improve the effectiveness, output, and environmental friendliness of agricultural methods.

7.2 Benefits of IoT integration

Integrated systems optimize agricultural operations, minimizing the time and manpower needed to carry out tasks such as planting, irrigation, and harvesting. Real-time monitoring and automation facilitate accurate resource management, reducing waste, and optimizing inputs [116]. Through the employment of data-driven insights and decision-making tools powered by artificial intelligence, farmers may enhance their workflows, optimize the utilization of equipment, and boost operational efficiency throughout the entire farm [117].

Productivity: Integrated systems improve production by providing farmers with precise and timely information, allowing them to make well-informed decisions. Precision agricultural technology, such as machinery commanded by GPS and automated irrigation systems, guarantees that crops are provided with the appropriate inputs at the correct moment. This results in enhanced yields and increased crop quality [118].



Fig. 7 Precision agriculture implementation: **a** tillage [112], **b** spraying [113], **c** planting [114], **d** harvesting [115]

Automation technologies, such as robotic harvesters and automated spraying systems, enhance the efficiency and precision of crop harvesting. This enables farmers to increase their harvesting speed and accuracy, resulting in reduced post-harvest losses [119].

Sustainability: Integrated systems enhance sustainability by reducing environmental harm and maximizing resource utilization. Precision agriculture techniques, such as the application of fertilizers and pesticides at varying rates, effectively mitigate chemical runoff and soil erosion, hence safeguarding soil health and water quality [120]. Automation technologies decrease the dependence on human labor, thereby addressing the labor shortage problems encountered by the agriculture business and enhancing the safety and welfare of farmworkers. Furthermore, integrated systems enhance the long-term sustainability of farming operations by efficiently utilizing resources and minimizing waste [121].

Cloud computing plays a vital role in agricultural data management because it provides scalable, efficient, and real-time solutions that improve precision agriculture techniques. Farmers can enhance resource control, productivity, and sustainability in their operations through the utilization of cloud-based technologies.

8 Challenges and limitations

8.1 Technical challenges

Sensor accuracy and reliability: Although there have been improvements in sensor technology, it is still difficult to ensure accuracy and reliability, particularly in challenging agricultural conditions. Various factors, such as changes in temperature, variations in soil conditions, and the deterioration of sensors, might have an impact on the quality and consistency of data [122]. Ensuring the accuracy and functionality of sensors

through calibration and maintenance is crucial, but it may be a burdensome and expensive task for farmers, especially those involved in large-scale operations. Maintaining the precision and durability of sensors necessitates continuous monitoring and regular recalibration [123].

Data integration and interoperability: The integration of data from multiple sources, such as different sensors and manufacturers, involves challenges related to the standardization and compatibility of the data. Non-compatibility between data types and communications protocols is a hindrance to the free exchange and integration of data [124]. Agricultural data frequently originates from several systems, encompassing machinery, meteorological stations, and satellite photography. To connect various systems and combine data into a single platform for analysis and decision-making, it is necessary to have strong data management and integration solutions [125].

8.2 Economic and scalability issues

High initial investment costs: Implementing precision agriculture technologies entails a significant upfront investment in hardware, software, and infrastructure. The cost barrier can discourage small and medium-sized farmers from adopting advanced technologies, thus limiting their capacity to benefit from precision farming [126]. Farmers with limited financial resources may find it difficult to afford the expensive sensors, drones, GPS devices, and other precision agriculture instruments. In addition, continuous costs associated with data subscription services, software upgrades, and maintenance place an additional financial burden [127].

Scalability and adaptability: The precision agriculture technologies that are developed for commercial large-scale farms might not be easily scalable or directly applicable to subsistence or smallholder farming conditions. The complexity of the technology and cost of adopting and maintaining advanced technologies might outweigh the potential benefit for smaller-scale businesses [128]. Scalability issues may also arise in terms of the processing and management of the data. With the rising volume of agricultural data, farmers and service providers are facing challenges in scaling their infrastructure and analytical capabilities to handle large datasets effectively [129].

8.3 Data privacy and security concerns

Data ownership and control: The ownership and control of agricultural data created by precision agriculture technologies, such as sensor readings, field maps, and yield estimates, are sources of concern. Due to concerns regarding privacy infringement or potential competitive disadvantages, farmers may exhibit hesitancy in sharing their data with service providers or third-party platforms [130]. Data ownership agreements between farmers and technology providers may suffer from a lack of transparency or clarity, resulting in ambiguity regarding the rights to the data and its potential uses or monetization. The lack of clarity in this situation might erode trust and hinder cooperation in projects involving the exchange of data.

Cybersecurity risks: Cybersecurity issues pose risks to precision agriculture systems with data breaches, unauthorized access, and cyber attacks. Precision farming is based on networked devices and cloud platforms, which are susceptible to cyber attackers seeking to disrupt operations or steal sensitive information [131]. The vulnerabilities in network security, lack of robust encryption mechanisms, and inadequate data protection

measures make agricultural data susceptible to the dangers of theft or tampering [132]. Any violation of data privacy or integrity could result in significant repercussions for farmers, impacting their ability to compete and damaging their brand [133].

Precision farming technologies can influence agricultural operations, but they are also fraught with challenges and hindrances. To be able to take full benefit of precision farming and to provide its applicability on a mass scale and in an equitable manner in agricultural fields, it is essential to solve technological, economic, and security problems.

8.4 Comparative analysis and trade-offs in technology

In an attempt to better comprehend the adoption and performance of new technologies in precision agriculture, the various technological strategies need to be compared more especially in areas of real-time decision support, data management, and system structure.

Edge vs. Cloud Computing: Cloud computing has become essential for processing and storing vast amounts of agricultural data due to its scalability, centralized control, and seamless integration with advanced AI models. However, it depends greatly on high data transmission latency and stable internet connection, which can hinder the real-time responsiveness in rural field conditions [134]. On the other hand, edge computing handles data locally at or near the point of origin (e.g., on-farm hardware or gateways), with greatly minimized latency and facilitating timely decision-making for timely use cases such as pest detection or irrigation management [135]. While edge computing enhances autonomy and offline functionality, it remains constrained by limited processing power and storage compared to cloud platforms.

Shortcomings of Present Technologies: Despite notable advancements, current agricultural technologies face several persistent challenges. IoT sensors often suffer from accuracy limitations, calibration drift, and high maintenance demands, especially under harsh field conditions. Machine learning algorithms employed for crop prediction or disease detection require large, labeled datasets, which are not always readily available in various agro-climatic environments [136]. Similarly, robotics and autonomous systems encounter difficulties navigating unstructured terrains and entail significant initial costs, limiting their accessibility and adoption among smallholder farmers.

Trade-Offs in System Adoption: Adopting a particular agricultural system often involves balancing trade-offs among cost, scalability, precision, and usability. For instance, autonomous tractors offer high accuracy and labor efficiency but are capital-intensive and may present operational complexity. Similarly, cloud-based AI analytics provide integrated, data-driven insights but raise concerns about data security and require constant internet connectivity [137]. Therefore, farmers and agritech stakeholders must carefully consider factors such as operational scale, digital literacy, infrastructure availability, and long-term return on investment (ROI) when selecting appropriate technologies.

9 Future directions and research opportunities

9.1 Emerging technologies

5G and edge computing: The implementation of 5G networks and progress in edge computing present favorable prospects for improving connectivity and data processing capacities in the field of agriculture. 5G networks enable fast and responsive

connectivity, allowing sensors, machines, and cloud platforms to transmit data in real-time with minimal delay [138]. Edge computing architectures, which include processing data close to the data source, effectively decrease latency and bandwidth demands. This, in turn, facilitates quicker decision-making and enhances resource management efficiency on the farm.

Blockchain technology: Blockchain technology can enhance transparency, traceability, and trust in agricultural supply networks. Blockchain technology offers immutable and decentralized ledgers, which guarantee the integrity and security of data about crop production, certification, and distribution. Blockchain-enabled smart contracts automate transactions and agreements, simplifying interactions among farmers, suppliers, distributors, and consumers while reducing the likelihood of fraud or disputes [139, 140].

Additive manufacturing and customization: The agricultural business has been looking into the use of 3D printing technology to develop customized equipment and products that are designed to meet specific requirements. This can result in enhanced efficiency and productivity in agricultural operations, along with cost reductions for farmers. In addition, precision agriculture methods, such as the use of drones and customized 3D printed sensors, can offer immediate solutions to availability and maintenance issues. This allows farmers to become self-reliant and not rely on the availability of specific replacement components. In summary, the incorporation of technology in agriculture has the potential to greatly improve the business and effectively tackle important issues like food security and sustainability [141].

9.2 Potential improvements in sensor technology and AI algorithms

Enhanced sensor capabilities: Further progress in sensor technology, including size reduction, enhanced sensitivity, and expanded functionality, will result in more adaptable and economical sensing systems for the agricultural sector. Upcoming sensors may provide enhanced precision, resilience, and energy efficiency, broadening their usability in many settings and circumstances. The use of new sensing methods, including hyperspectral imaging, LiDAR, and spectroscopy, alongside conventional instruments, can offer a thorough understanding of soil quality, crop function, and environmental circumstances. This integration allows for more accurate and comprehensive decision-making [142].

AI and machine learning innovations: Continuing research in AI and ML algorithms is anticipated to enhance predictive modeling, anomaly detection, and decision support systems for agriculture. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are advanced deep learning approaches that excel in tasks such as image analysis, pattern recognition, and time-series forecasting. Explainable AI (XAI) methods strive to improve the comprehensibility and clarity of AI models, allowing farmers and stakeholders to grasp the logic behind automated suggestions and choices. Interdisciplinarity between data scientists, agronomists, and domain experts will facilitate the development of AI solutions appropriate for agricultural concerns [143].

9.3 Long-term sustainability and environmental impacts

Climate-resilient agriculture: The impact of climate change on agricultural output and food security is substantial, requiring the implementation of farming systems that are adaptable to climate fluctuations [144]. Utilizing precision agriculture technologies, in

conjunction with climate modeling and forecasting techniques, enables farmers to adjust to evolving environmental circumstances and minimize the potential hazards linked to severe weather occurrences. Investigating climate-smart agricultural techniques, such as conservation tillage, cover cropping, and agroforestry, will enhance soil health, preserve water resources, and store carbon, promoting long-term sustainability and resilience in agriculture [145].

Environmental monitoring and management: Integrated environmental monitoring systems, which utilize IoT sensors, satellite imaging, and remote sensing technologies, allow for the immediate evaluation of ecosystem health and biodiversity [146]. These systems enable the proactive management of natural resources, such as water resources, soil fertility, and wildlife habitats. The implementation of precise conservation techniques, such as focused reforestation, wetland rehabilitation, and riparian buffer zones, can effectively reduce the negative environmental effects of agriculture, improve ecosystem services, and support the preservation of biodiversity across large areas of land. The future of agricultural research and technology development is ready to tackle important challenges and possibilities in sustainability, productivity, and resilience. Through the utilization of developing technologies, enhancement of sensor capacities, and progression of AI algorithms, the field of agriculture can shift towards practices that are more efficient, fair, and environmentally sustainable [147].

10 Conclusion

Precision agriculture is a cutting-edge method of farming that combines technologies like sensors, cloud computing, IoT, AI, ML, and automation to make the agricultural system more efficient and sustainable. Research suggests that precision technologies have the potential to bring significant changes, with water savings of 20%–30% achieved using smart irrigation techniques and fertilizer usage reduced by up to 15% with variable rate application methods. In the same way, AI-based pest detection systems have had accuracy levels between 85 and 95% in pilot trials. While these statistics point to the promise of precision agriculture, actual gains are contingent on aspects like crop variety, regional climate, infrastructure, and deployment scale.

In the future, continued advancements in technologies such as 5G, edge computing, and blockchain can potentially provide better connectivity, quicker data processing, and increased transparency in agricultural supply chains, though their large-scale implementation and effect on farm-level productivity need to be substantiated. Improvements in sensor technologies and AI algorithms are anticipated to enhance the accuracy of decision-support tools, with some field tests reporting a 10% yield increase when integrated systems are used. By integrating these technologies with climate-resilient methods and precision conservation approaches, we can ensure long-term sustainability, minimize environmental degradation, and contribute to food security. To reap the maximum benefits of precision agriculture, it is important to resolve the concerns of inclusion, accessibility, and ethics in a way that smallholder and marginalized farmers are not disadvantaged during the technological shift.

Author contributions

Mrutyunjay Padhiary Writing—original draft, writing—review & editing, visualization, resources, methodology, investigation, conceptualization. Avinash Kumar: Writing—review & editing, supervision, software, resources. Laxmi Narayan Sethi: Data curation, formal analysis, resources.

Data availability

No datasets were generated or analysed during the current study.

Declarations**Ethics approval and consent to participate**

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Consent for publication

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